EXP NO: 11 EXPERIMENT TO UNDERSTAND KNN ALGORITHM FOR A GIVEN DATA SET

AIM:

Understand the KNN algorithm for the dataset given.

PYTHON CODE:

```
import numpy as np
import pandas as pd

df = pd.read_csv('Iris.csv')
df.info()
df.variety.value_counts()
df.head()
features = df.iloc[:,:-1].values
label = df.iloc[:,4].values
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
xtrain, xtest, ytrain, ytest = train_test_split(
   features, label, test_size=.2, random_state=1
)
model_KNN = KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain, ytrain)
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))
from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
0.9583333333333334
1.0
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.96      0.94      0.95        50
   Virginica       0.94      0.96      0.95        50

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150
```

RESULT:

Thus the code is executed successfully.